Supplementary Materials

Age Weighting Procedure

Weights are applied to each couple such that the joint age distribution of extant couples in a given year will match the joint age distribution of newlyweds in that same year. The weight w_{ij} for a couple with the husband in age group i and the wife in age group j is given by:

$$w_{ij} = \frac{p_{ij}^{(N)}}{p_{ij}^{(E)}}$$

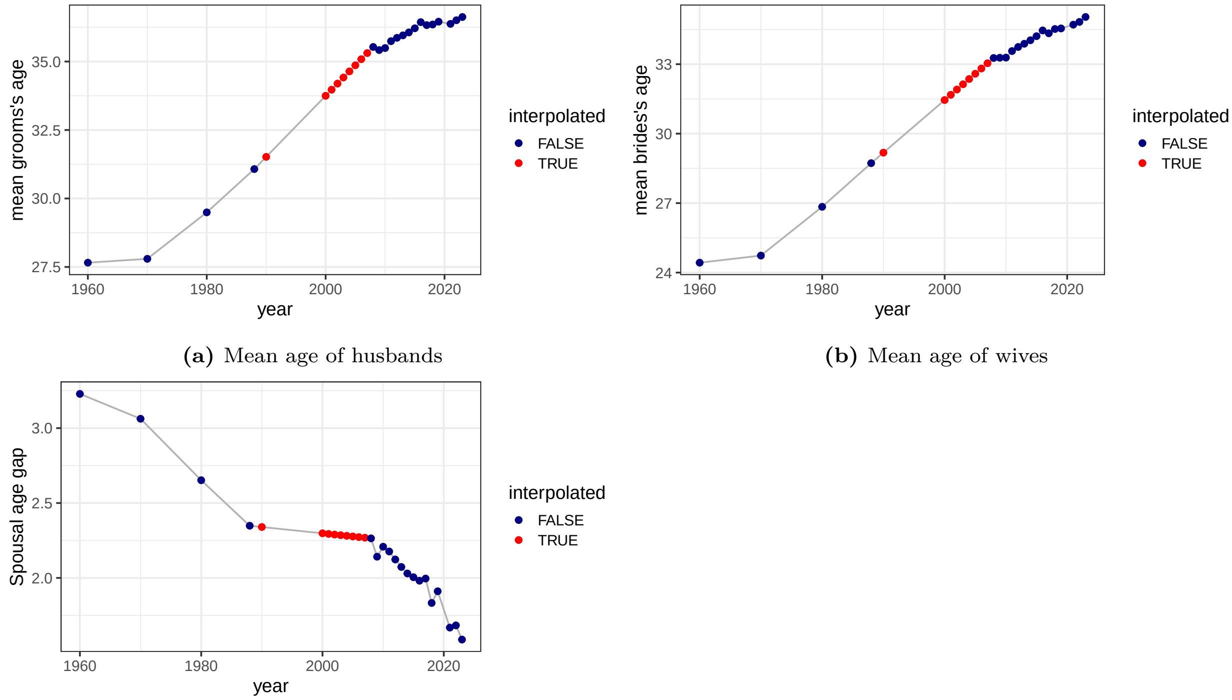
where w_{ij} is the weight for couples where the husband belongs to age group i and the wife belongs to age group j, $p_{ij}^{(N)}$ is the proportion of newlywed couples belonging to the joint age group, and $p_{ij}^{(E)}$ is the proportion of extant couples in the Census/ACS data source belonging to the joint age group.

To determine $p_{ij}^{(N)}$ for the entire time period, I rely upon two sources. From 2008 onward, ACS data includes information on marital timing for all current marriages and therefore I can estimate this value directly from the data by taking all couples married a year or less. Prior to 2008, I rely upon US vital statistics reports on marriage for 1960, 1970, 1980, and 1988 (National Center for Health Statistics 1960, 1970, 1980, 1988). Each of these sources provides tables of the joint age distribution of newlyweds, originating from marriage records. However, each source has idiosyncrasies that present challenges.

First, the sources use different age groupings. I therefore use an age grouping that can be made consistent across sources by summing smaller groups in more detailed sources. The age grouping I use for both husbands and wives is under 20, 20-24, 25-29, 30-34, 35-44, 45-54 and 55 or over. Second, some data sources use tables that are disaggregated further, requiring me to sum results across multiple tables. The 1960 data are split by the previous marital status of brides (single, divorced, widowed). The 1988 data are split by the race of newlyweds into White, Black, and Other and exclude multiracial couples.

These data sources provide me with the joint age distribution up to 1988. After 1988, these reports were no longer produced. To calculate the joint age distribution for Census 1990, Census 2000, and the ACS from 2001-2007, I linearly interpolate the joint age distribution from 1988 to 2008. Specifically,

$$p_{ij}^{(t)} = (1 - \alpha) * p_{ij}^{(1988)} + \alpha * p_{ij}^{(2008)}$$



(a) Mean age of husbands

(b) Mean age of wives

(c) Spousal age gap

Figure S1. Summary measures of age distribution of spouses, interpolated and non-interpolated data

Where $p_{ij}^{(t)}$ denotes the proportion of couples in which the husband belongs to age group i and the wife belongs to age group j in year t. The interpolation weight, α, is given by:

$$\alpha = \frac{t_i - 1988}{2008 - 1988} = \frac{t - 1988}{20}$$

Figure S1 examines the mean age of grooms and brides, and the age difference between them across the time period with years of interpolation highlighted in red. The interpolated results are smoother than the later time period which relies on single years of ACS data, but otherwise produces a reasonable interpolation across the years of missing data.

As a further check, Figure S2 shows heat maps of the full joint distribution of newlywed age for the years of the interpolated distributions. The years of 1988 and 2008 provide actual data. The figure shows a smooth transition in the intervening time period in the joint age distribution, as the ages of grooms and brides both increase and the age gap between them declines slightly.

Comparison of Age-Weighted and Duration-Specific Results

Age-weighting the frequency counts of extant marriages for log-linear models to match the joint age distribution is superior to restricting the models to certain age ranges of husbands and/or wives. However, it is unlikely to exactly match the actual duration-specific data. First, the extant marriage data is affected by marriage dissolution and that information cannot be

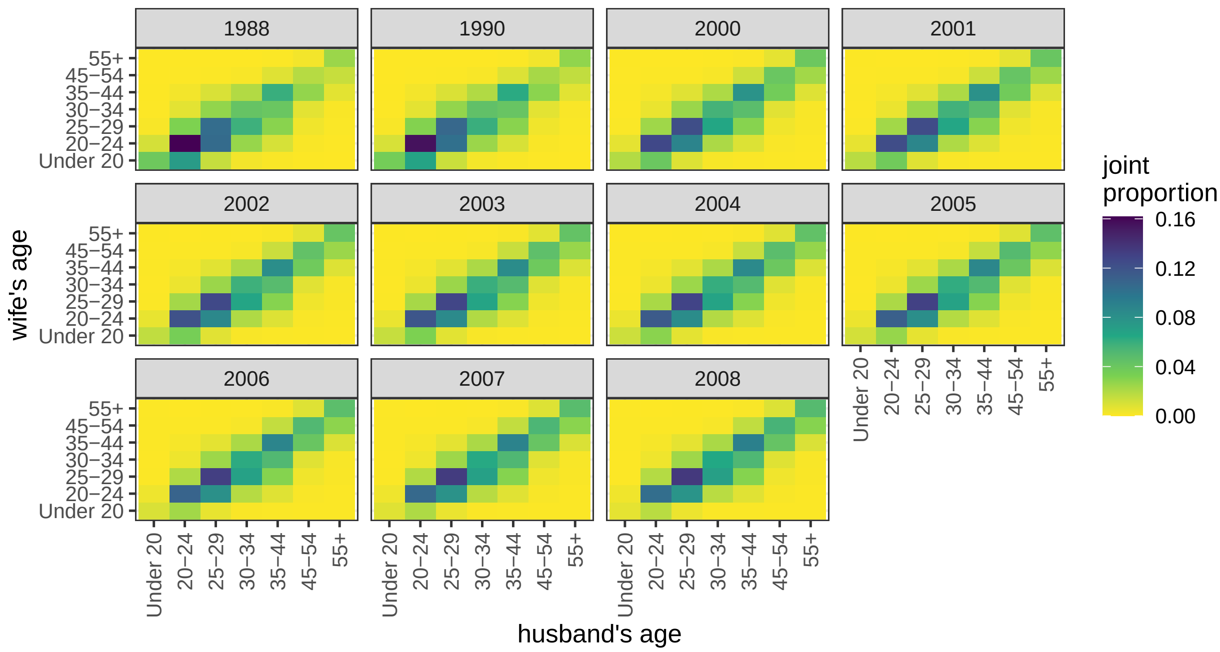


Figure S2. Heat map of joint distribution of spouse's age, real and interpolated years

recovered. Second, although changes in the likelihood of interracial marriage by birth cohort are accounted for within the age weighted approach, period changes that identify marriage cohorts are not. For example, lets compare two marriages between fifty year old partners, one of which was formed this year and the other which was formed 25 years ago. Birth cohort related effects on the likelihood for interracial marriage (e.g. internalized preferences for partners of a given race) are the same across both sets of couples. However, period specific changes in that likelihood (e.g. a more liberal atmosphere, greater opportunities to meet cross-race partners) will differ for the two cases, and thus averaging out their likelihoods will underestimate the current likelihood.

In both cases, the direction of the bias arising from using age-weighted extant data is predictable. Marriage dissolution is more likely for interracial couples (Bratter and King 2008) and the general period effects over this time period indicate a more liberalizing environment with regard to interracial marriage. Therefore, I expect that models with actual duration-specific data will find a somewhat higher likelihood of interracial marriage than age-weighted extant data.

For some years of the time period, I have data to examine the direction and strength of the bias directly. From 1960 to 1980 and after 2007, I have information on the duration of marriage that I can use to restrict the sample to individuals married in the last year. For the 1960 to 1980 period, this information is only provided for first marriages. Figure S3 compares results based on duration data to the full results that use age weighting to make the sample representative of the newlywed population.

As expected, the duration-specific results produce slightly higher odds of interracial marriage for most groups. The degree of difference is higher in the early period between 1960 and 1980.

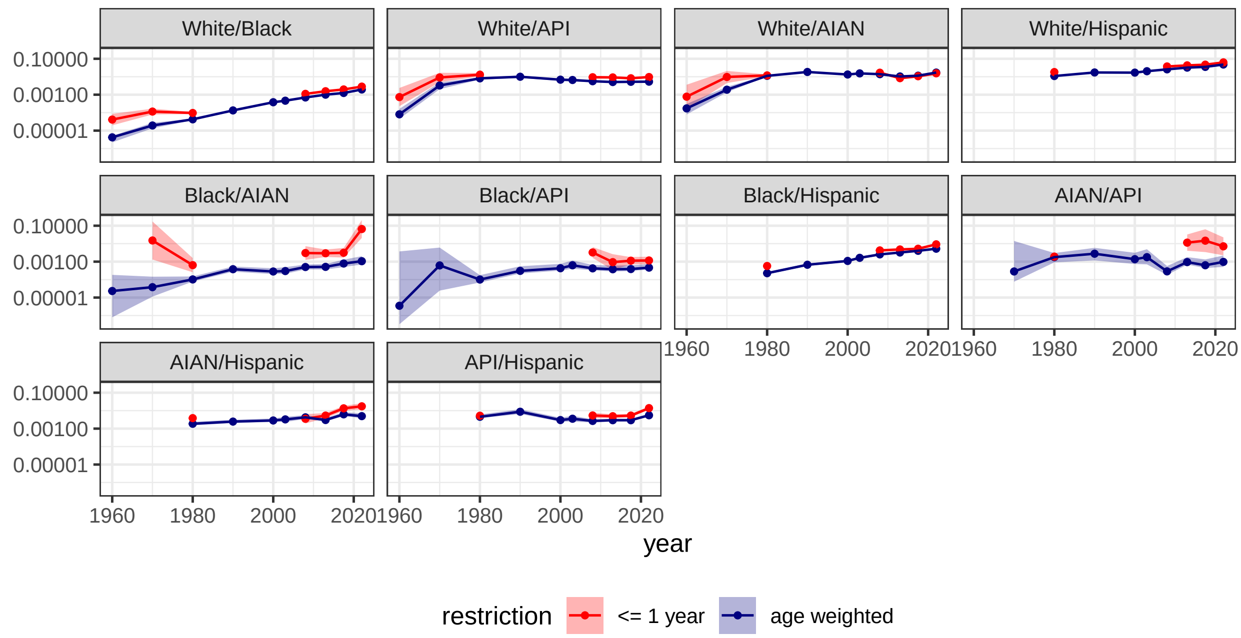


Figure S3. Comparison of odds of racial exogamy for single race pairings that involve the Hispanic or API by age weighted results and newlyweds (within a year). Estimates are derived from log-linear models that control for state racial composition. Age weighted results are weighted to reflect the joint age distribution of spouses. 83% confidence bands shown.

However, the duration data here is limited to first marriages which may also lead to some discrepancy with the age weighted data which are for all marriages. Overall, the discrepancy between the two approaches is minor and the trends across the two approaches are similar.

Inclusion of Foreign-Born Population

The main results for the analysis excludes unions involving foreign-born individuals. For data without marriage duration information, it is impossible to determine whether such unions were formed in the US or elsewhere. Furthermore, foreign-born individuals will likely have a stronger attachment to marrying someone of the same ethnonational group even when marrying in the US. For these reasons, including foreign-born individuals will likely lower the overall likelihood of racial exogamy.

Figure S4 compares the result for the main analysis, which only includes the native born, to results that include the foreign-born population. I experiment with only including the "1.5" generation, which consists of individuals who arrived in the US before age 18, and the entire foreign-born population. The results show that inclusion of the foreign-born population depresses the odds ratio of interracial marriage for all pairings that involve an API or Hispanic partner. As expected, the results for White/Black, White/AIAN, and Black/AIAN marriage are unaffected by the inclusion. The results look similar regardless of whether the entire foreign-born population is included or only the 1.5 generation. Although the rates are lower when including the foreign-born population, the overall trends are similar.

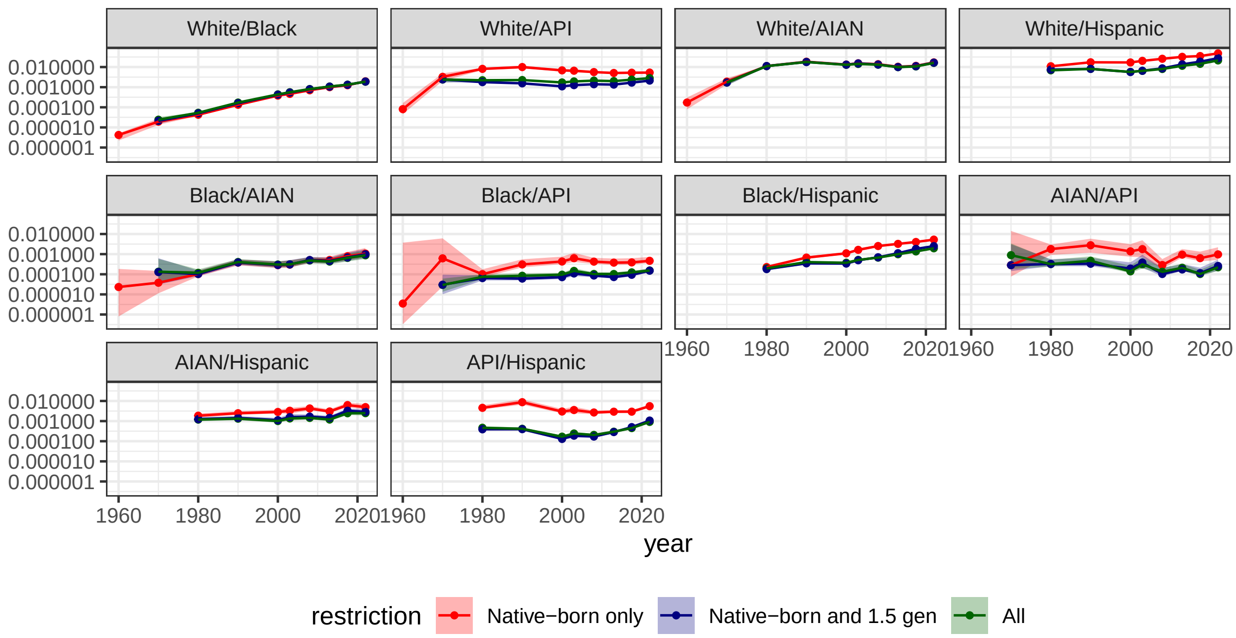


Figure S4. Comparison of odds of racial exogamy for single race pairings that involve the Hispanic or API by restriction on foreign-born population. Estimates are derived from log-linear models that control for state racial composition and are weighted to reflect the joint age distribution of spouses. 83% confidence bands shown.

References

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